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Analysis on Best Model Performance

Following midterm feedback, data leakage was eliminated by strictly executing a 70% Train /15% Validation / 15% Test stratified split *prior* to fitting any imputation, scaling, or SMOTE operations. Missing numeric values were imputed using training set medians, and domain-specific climate features were added (diurnal temperature range `TEMP_RANGE`, apparent stress proxy `HEAT_INDEX_PROXY`, cyclical month transformations, and soil-air thermal differences).

Model Performance & Analysis

Best Model: The Bayesian Weighted Ensemble achieved the top performance across all evaluation metrics with a ROC-AUC of 0.991, F1-Score of 0.940, and Accuracy of 0.964. Individual tree-based gradient boosting (XGBoost) proved to be the strongest standalone classifier (ROC-AUC 0.985). Tree ensembles excel on climate data because they handle complex non-linear interactions among meteorological features without requiring linear separability.

Worst Model: K-Nearest Neighbors (KNN) produced the lowest metrics (F1-Score 0.806, Accuracy 0.871). KNN struggled due to distance metric distortion caused by high dimensionality following feature engineering and categorical encoding.